

[DHS/2022/Q3]

Task 3

Save all your folders and files for this task in a folder named

TASK3_<name>

You will be creating a NDP ticketing web application `main.py` for applicants to check their NDP balloting results.

Task 3.1

You will be using a SQLite database for the program.

Create a database `ballot.db` with data from the following tables `Names` and `Results`. This data is also provided in `Names.csv` and `Results.csv`.

Results

nric	group_id	ballot_result
T0663758C	NULL	yes
S7829998G	4	yes
T0487877S	4	no
S8735513R	NULL	yes
T0254372E	NULL	yes
S9331280R	NULL	no
S9981511E	4	yes
S7851994I	6	no
S6450103G	6	yes

Names

nric	names
T0663758C	Farah Yusoff
S7829998G	Terry Sathyalingham
T0487877S	Lin Wee Kiat Micheal

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S8735513R	Randal Khiatani
T0254372E	Chen Chi Chien
S9331280R	Natalie Lee
S9981511E	Nayeli Sng Hui Hoon
S7851994I	Cherilyn Leong
S6450103G	Emma Wijeysingha

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Task 3.2

Provide SQL statements in `task3_2.sql` to

- retrieve the `ballot_result` of T0663758C
- retrieve the `name` and `ballot_result` of the other applicants who are in the same group as S9981511E (assume you do not know the `group_id` of S9981511E when writing this query)

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Task 3.3

Create two HTML template files:

`home.html` will receive an NRIC as input:

NRIC:

Upon clicking the submit button of `home.html`, your program should use Flask to direct the browser to display the balloting results in `dashboard.html` for the NRIC entered. Invalid NRICs, NRICs which are not in the database, will be shown a page stating “Invalid NRIC”.

`dashboard.html` displays the applicant’s balloting results, along with a table of results of other applicants who applied to be in the same group, if any:

Your NDP balloting results: yes

Results of members in your group:

Name	Balloting Results
Terry Sathyalingam	yes
Lin Wee Kiat Micheal	no

Build the above functionalities and test the web application on a local server.

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Task 3.4 (Bonus question)

Implement the following:

- The border styling of the table of dashboard.html matches the above image in Task 3.3.
- Create `styles.css` and use selectors to ensure all balloting results (group and individual) will be in red.
- For the case of an individual with no assigned grouping, the browser should display “No group data” in the fields:

Your NDP balloting results: yes	
Results of members in your group:	
Name	Balloting Results
No group data	No group data